

Lecture 11: Differentials; chain rule

More tools to study functions

Differentials: $y = f(x)$

Implicit differentiation: $dy = f'(x) dx$

Example: $y = \sin(x)$

$x = \sin(y) \Rightarrow \cos(y) dy = \frac{1}{\cos(y)} = \frac{1}{\sqrt{1-\sin^2(y)}}$

Total differential: $f(x, y, z)$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

Important: df is NOT Δf

Can do: ③ Explore how changes in x, y, z affect f

② procedure for small variations $\Delta x, \Delta y, \Delta z$ to get approx formula $\Delta f \approx \Delta_x f + \Delta_y f + \Delta_z f$

③ Divide by something like dt to get a rate of change

$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$ when $x = x(t), y = y(t), z = z(t)$

Chain Rule

Why is this valid?

Let Attempt: $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$ $dx = x'(t) dt$ $dy = y'(t) dt$ $dz = z'(t) dt$

$\therefore df = \frac{\partial f}{\partial x} x'(t) dt + \frac{\partial f}{\partial y} y'(t) dt + \frac{\partial f}{\partial z} z'(t) dt$ Divide by $dt \Rightarrow$ Chain rule

Bottom: $\Delta f \approx \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \frac{\partial f}{\partial z} \Delta z$ (in a time Δt)

When $\Delta t \rightarrow 0$, $\frac{\Delta f}{\Delta t} \rightarrow \frac{df}{dt}$, $\frac{\Delta x}{\Delta t} \rightarrow \frac{dx}{dt}$ so $\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$ $\therefore$ becomes $\therefore$ limit $\Delta t \rightarrow 0$

Example: $w = x^2 y + z$, $x = t$, $y = e^t$, $z = \sin t$

Chain Rule: $\frac{dw}{dt} = 2xy \frac{dx}{dt} + x^2 \frac{dy}{dt} + 1 \frac{dz}{dt} = 2t e^t + t^2 e^t + \cos t = 2t e^t + t^2 e^t + \cos t$

Substitute: $w = x^2 y + z$

$w(t) = t^2 e^t + \sin(t)$

Application: Justify (product quotient) rules

$\frac{dw}{dt} = 2t e^t + t^2 e^t + \cos(t)$

$f = uv$ $u = u(t)$, $v = v(t)$ $\frac{d(uv)}{dt} = \frac{\partial}{\partial t} \frac{d}{dt} = v \frac{du}{dt} + u \frac{dv}{dt}$

$g = \frac{u}{v}$ $u = u(t)$, $v = v(t)$ $\frac{d(\frac{u}{v})}{dt} = \frac{1}{v} \frac{du}{dt} + \left(\frac{-u}{v^2} \right) \frac{dv}{dt} = \frac{u'v - uv'}{v^2}$ Use chain rule with more variables

$w = f(x, y)$, where $x = x(u, v)$ $y = y(u, v)$ Question: $\frac{dw}{du}$, $\frac{dw}{dv}$ in terms of $\frac{dw}{dx}$, $\frac{dw}{dy}$, x_u , x_v , y_u , y_v

$dw = \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy$
 $= \frac{\partial w}{\partial x} (x_u du + x_v dv) + \frac{\partial w}{\partial y} (y_u du + y_v dv)$
 $= \left(\frac{\partial w}{\partial x} x_u + \frac{\partial w}{\partial y} y_u \right) du + \left(\frac{\partial w}{\partial x} x_v + \frac{\partial w}{\partial y} y_v \right) dv$

$\frac{dw}{du} = \frac{\partial w}{\partial x} \frac{dx}{du} + \frac{\partial w}{\partial y} \frac{dy}{du}$

$\frac{dw}{dv} = \frac{\partial w}{\partial x} \frac{dx}{dv} + \frac{\partial w}{\partial y} \frac{dy}{dv}$

Example: polar coordinates $f(x, y)$

$x = r \cos \theta$ $y = r \sin \theta$ $\frac{dw}{dr} = \frac{\partial w}{\partial x} \frac{dx}{dr} + \frac{\partial w}{\partial y} \frac{dy}{dr} = \frac{\partial w}{\partial x} \cos \theta + \frac{\partial w}{\partial y} \sin \theta$

